AI-Based Intelligent Road Damage Detection, Severity Assessment, Repair Prioritization and Predictive Road Maintenance System

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Road damage is a problem for road officials as fixing issues often happens after problems are already visible. This way of dealing with problems comes after they happen which can make repair costs higher make roads less safe and make the roads not last long. Even though current ways to check roads can find problems and see how roads are doing it is still hard to get enough help for planning maintenance before problems get worse.

This study is about predicting how roads will get worse in the future and figuring out when it is best to do maintenance before it's too late. Records of road conditions past maintenance work how much traffic there is, the weather and current road conditions can show patterns of how roads get worse.. Many current methods only look at current damage and don't use these patterns to make predictions or suggest ways to prevent problems.

To solve this an AI-based part for predicting road maintenance is suggested. This part will guess how roads will be in the future, how long roads will last what maintenance is needed and the best ways to do work. Data will be. Looked at using Pandas and NumPy while different machine learning methods like Scikit-learn models, XGBoost, CatBoost and LightGBM will be tested and compared. The development will include getting data cleaning it creating features, training models making predictions and coming up with recommendations.

The models will be tested using the methods for prediction like MAE, RMSE, R^2, precision, recall and F1-score depending on what is needed. The new part will work as the part that predicts and plans in the road maintenance system using information from other parts of the system to help make better and earlier decisions, about maintenance.

Keywords: Predictive Road Maintenance, Machine Learning, Road Deterioration, Remaining Road Life, Preventive Maintenance

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We are thankful to the staff of the university for giving the knowledge, resources and support needed to carry out this research. We also thank the researchers and organizations whose publications, datasets and technical resources gave background for our study on road condition assessment, deterioration prediction and maintenance planning.

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CONTENTS

DECLARATION 2

ABSTRACT 3

1.1 Background and Context 11

2 .Literature Review 12

1.3 Research Gap 14

2.2 Specific Objectives 15

3.4 Evaluation Metrics 19

3.7 Tools and Platforms with Justification 21

4. HIGH-LEVEL SYSTEM ARCHITECTURE 21

4.1 Overview of the Proposed Solution 21

4.2. Individual Component Architecture 22

4.3 Overall Integration 23

4.4 Dependencies 23

Dependency 23

Purpose 23

Historical Road Data 23

Provides detected defects and damage information required for deterioration analysis. 23

Road Characteristics 23

Provides information such as road type, age, and other relevant attributes 23

Traffic Data 23

Helps identify the effect of traffic conditions on road deterioration. 23

Environmental Data 23

Provides factors such as rainfall and temperature where available 23

Maintenance History 23

Provides information about previous maintenance activities and their timing. 23

System Integration 23

Allows Component 4 to receive required inputs and provide predictions to the complete system. 23

4.5 Individual Responsibility Boundary 24

5. USER REQUIREMENTS 24

5.1 Stakeholders 24

5.2 Evidence Collection 24

5.3 Functional Requirement 25

5.4 Non-Functional Requirement 25

6.0 COMMERCIALIZATION PLAN 25

6.1 Target Market 25
6.2 Customer Persona 25
6.3 Value Proposition 26
6.4 Revenue Model 26
6.4.1 System Deployment Fee 26
6.4.2 Software Subscription 26
6.4.3 Maintenance and Support 26
6.4.4 Analytics Service 26
6.4.5 Proposed Revenue Structure 26
6.5 Cost Estimation 27
6.6 Pricing Strategy 27
6.7 Competitive Advantage 27
6.8 Intellectual Property Considerations 27
7.0 BUDGET AND JUSTIFICATION 27
7.1 Hardware costs 27
7.2 Software Tools 28
7.3 Human Effort Cost 28
8.0 WORK BREAKDOWN STRUCTURE 29
9.0 GANTT CHART 30
10. REFERENCES LIST 31
11. APPENDICES 32

LIST OF TABLES

Table 1- Table of evaluation matrices…… 17
Table 2-Table Tools and Platforms with Justification……..20
Table 3-Table of Dependencies…… 24
Table 4-Table of Software Tools Hardware costs.....33
Table 5-Table of Hardware costs 33
Table 6-Table of Human effort cost 34

LIST OF FIGURES

Figure 1- High level system Architecture..... 21
Figure 2- Overall Integration..... 23
Figure 3- Individual Responsibility Boundry..... 24
Figure 4- Work breakdown structure..... 34
Figure 5- Gantt chart..... 35

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CV	Computer Vision
DSR	Design Science Research
F1-Score	Harmonic Mean of Precision and Recall
FR	Functional Requirement
IoT	Internet of Things
ML	Machine Learning
NFR	Non-Functional Requirement
OpenCV	Open-Source Computer Vision Library
SDG	Sustainable Development Goal
UI	User Interface
WBS	Work Breakdown Structure

LIST OF APPENDICES

Appendix	Title
Appendix A	Research Project Gantt Chart
Appendix B	Work Breakdown Structure (WBS)
Appendix C	Proposed System Architecture / System Overview
Appendix D	Research Methodology Workflow
Appendix E	Data Collection and Dataset Description
Appendix F	Sample Dataset Attributes / Data Dictionary
Appendix G	Machine Learning Model Development Framework
Appendix H	Evaluation Metrics and Validation Plan
Appendix I	Ethical Clearance and Data Privacy Considerations
Appendix J	Individual Component Contribution Breakdown

1. INTRODUCTION

1.1 Background and Context

Road infrastructure is very important for transportation for business activities and for getting to important services.. Road surfaces slowly get worse because of heavy traffic, weather getting older, water problems and past damage. If these changes are not noticed and fixed in time small problems can become bigger road failures. This can make fixing roads more expensive make the roads less safe and cause trouble for people who use the roads.

In road maintenance methods decisions are mostly based on how a road looks right now or on damage that is already visible. Checking and looking at the condition of a road can help people in charge understand what is happening with the road.. Knowing what the road looks like now does not always give enough information to decide when a road section will need fixing. Because of this maintenance can be reactive meaning action is taken after the road is already in shape.

More and more data about road conditions and other related information is becoming available. This gives a chance to make the process better. Records of how roads have gotten worse over time past work done on roads, traffic data, weather, road features and current measurements can all be looked at together to find patterns that show how roads get worse. Machine learning is helpful here because it can study the relationships in data and use those relationships to guess what might happen in the future.

This research therefore looks at road maintenance not as a way to find damage that is already there but also as a way to plan for the future. It needs to understand how road conditions might change over time. In the system that is being suggested Component 4 focuses on this part of predicting what will happen. It uses the information that comes from all parts of the system to predict how the road will look in the future how long the road will last, when maintenance will be needed and to suggest ways to fix things before they get worse. This helps the overall plan move from pointing out road problems to helping plan, for road maintenance earlier and better.

2 .Literature Review

Road maintenance has gradually moved from traditional inspection-based practices towards data-driven approaches that can support more timely maintenance decisions. Earlier methods mainly relied on visual inspections, pavement condition surveys, and engineering judgement to identify damaged road sections. Although these methods are still useful for understanding the present condition of a road, they provide limited support for predicting how that condition will change in the future. Recent studies have therefore explored machine learning and predictive modelling as methods for understanding road deterioration and supporting maintenance planning.

2.1 Road Condition and Deterioration Prediction

Road deterioration is influenced by several factors rather than a single condition. Traffic loading, pavement age, environmental conditions, road characteristics, previous maintenance activities, and existing defects can all contribute to changes in road condition over time. For this reason, historical and current road information can be valuable when estimating future deterioration.

Previous research has applied statistical and machine-learning techniques to identify relationships between these factors and changes in pavement condition. Predictive models can learn from previously observed road behaviour and estimate future condition values for similar road sections. This provides an advantage over approaches that only identify damage after it has already occurred.

However, the usefulness of a deterioration model depends strongly on the quality and relevance of its input data. Missing records, inconsistent measurements, and differences between road environments can reduce prediction reliability. Therefore, data cleaning, preprocessing, feature selection, and appropriate validation are important stages when developing a practical road deterioration prediction model.

2.2 Machine Learning for Road Maintenance

Machine-learning techniques have become increasingly relevant to road infrastructure management because they can analyse relationships among multiple road-related variables. Traditional regression techniques can provide useful baseline predictions, while ensemble learning methods can model more complex relationships in the data.

Algorithms such as Random Forest and gradient-boosting approaches, including XGBoost, LightGBM, and CatBoost, are particularly relevant when working with structured road datasets. These models can be trained using historical observations and evaluated to determine which approach provides the most reliable predictions for the available dataset. Instead of assuming that one algorithm will always perform best, comparing several candidate models provides a stronger basis for model selection.

For the proposed component, this comparison is important because the final model should provide useful predictions for maintenance planning rather than only achieve good

performance on training data. Appropriate preprocessing, feature engineering, model tuning, and testing with unseen data are therefore required.

2.3 Remaining Road Life and Future Condition

Knowing the present road condition is useful, but maintenance planning also requires an understanding of how long acceptable performance can be expected to continue. Remaining road life can provide an estimate of the period before a road section reaches a condition where maintenance or rehabilitation becomes necessary.

Research on pavement performance prediction shows that historical condition trends can be used to estimate future deterioration. When combined with relevant influencing factors, predictive models can support the estimation of future condition and possible maintenance periods. Such information can help road authorities identify sections that may require attention earlier, rather than waiting until severe deterioration is observed.

Nevertheless, predicting remaining life is challenging because deterioration does not occur at the same rate for every road. Differences in traffic, climate, construction, maintenance history, and available data must be considered when interpreting model outputs. Therefore, remaining-life predictions should be treated as decision-support information rather than an exact representation of future road behaviour.

2.4 Preventive Maintenance Planning

Preventive maintenance aims to address road deterioration before major failure occurs. Compared with waiting for severe damage, earlier maintenance can help preserve road condition and reduce the need for more extensive repairs. Effective preventive planning requires more than detecting a defect; it requires information about the likely future condition, expected maintenance need, and appropriate time for intervention.

Existing research demonstrates the value of predictive models for estimating pavement condition. However, a practical gap remains between producing a prediction and converting that prediction into information that can directly support maintenance decisions. A model may provide a future condition value, for example, without clearly indicating what that value means for maintenance planning.

Component 4 addresses this issue by connecting prediction with maintenance-oriented outputs. Predicted future condition and remaining road life will be interpreted together with relevant road information to forecast maintenance requirements and generate preventive maintenance recommendations.

2.5 Identified Research Gap

The reviewed approaches indicate considerable progress in road damage assessment and pavement condition prediction. Computer-vision-based systems can identify visible defects, while machine-learning models can estimate condition or deterioration using historical data. However, these capabilities are often studied separately.

A limitation for practical road maintenance is that identifying the current problem does not by itself indicate how quickly the road may deteriorate, when maintenance may be required, or what preventive action should be considered. Similarly, prediction results have limited operational value if they are not translated into understandable maintenance information.

Therefore, the identified gap is the need for an integrated predictive component that uses current and historical road information to support forward-looking maintenance decisions. Component 4 is proposed to address this gap by combining future road condition prediction, remaining road life estimation, maintenance forecasting, and preventive maintenance recommendations. Within the complete system, this component extends road assessment from identifying what is happening at present to providing information about what may happen next and how maintenance teams can respond earlier.

1.3 Research Gap

I see that road maintenance research has come a way. It now uses image processing, computer vision and machine-learning to spot road defects and check pavement condition. These tools are good, at finding cracks, potholes and worn surfaces.. Most of this work only looks at a road after it has already started to deteriorate.

I notice that other studies have tried to forecast road wear by using data. They show that machine learning can guess road condition but the guess is usually taken as the end result. In practice there is still a gap. Planners cannot just rely on a prediction; they need more. They want to know how much useful road life is left when maintenance will be needed and what preventive steps can be taken.

I find another problem: road condition prediction, remaining-life estimation and maintenance advice are often treated as jobs. Because of that it is hard to turn the data we have into one picture that helps planners act early.

Component 4 is our answer. It builds a maintenance tool that mixes current and past road data to estimate future road condition and how long the road will keep working. Those estimates will then predict when maintenance will be needed and give maintenance suggestions.

I think this component goes beyond just spotting existing road problems. It gives forward-looking information that helps maintenance authorities act before the road gets worse.

2. OBJECTIVE

2.1 Main Objective

The goal of this part of the research is to create a way to use intelligence to predict what will happen with roads. This approach will look at data about roads now. From the past. It will try to guess how roads will be, in the future. It will also try to figure out how long roads will last. It will try to predict what kind of work roads will need.. It will give ideas about the best ways to take care of roads before problems happen. This will help people plan road maintenance earlier and make choices.

2.2 Specific Objectives

1. To find out and look at the things that affect how roads get worse including how the road is now past records of how the road was, information about traffic things in the environment features of the road and any past work done on the road.

2. To get ready. Work with the data that is collected about roads by cleaning the data changing it into a better format creating new features and choosing the most important parts for making predictions.

3. To. Check different machine-learning models that can predict how roads will be in the future and figure out how long the road sections will last.

4. To create a way to predict when a road section might need work by using the predictions about the road condition and how long it will last.

5. To come up with ideas for taking care of roads before problems happen by using the predictions about how the road's doing and what work is needed helping to plan better and earlier maintenance in the whole system, for road work.

3. METHODOLOGY

3.1 Research Methodology / Design

This research will follow a data-driven approach to develop and evaluate the predictive maintenance component. The methodology is organised into stages beginning with the preparation of road-related data and ending with the generation and evaluation of maintenance predictions and recommendations.

1. Data Collection and Integration

The component will use current and historical road-related data. Relevant attributes may include road condition records, identified road defects, previous maintenance activities, road age and characteristics traffic-related information and environmental factors. Information generated by components of the complete system will also be considered where it is relevant to future deterioration prediction.

2. Data Preprocessing

The collected data will be examined for missing values, duplicate records, inconsistent formats and possible outliers. Numerical and categorical variables will be prepared in a form for machine-learning models. Pandas and NumPy will mainly be used for data cleaning, transformation and initial analysis.

3. Feature Selection and Engineering

Relevant features that may influence road deterioration will be identified from the dataset. Existing variables may also be. Combined to create meaningful features representing factors such as road condition trends, maintenance history, deterioration rate and exposure to traffic or environmental conditions.

4. Model Development

Several machine-learning approaches will be investigated than relying on a single algorithm. Scikit-learn-based baseline and ensemble models will be compared with gradient-boosting techniques such as XGBoost, CatBoost and LightGBM. The dataset will be separated into training and testing sets and model parameters will be adjusted using the training data.

5. Future Condition and Remaining Road Life Prediction

The selected model will be used to estimate how the condition of a road section may change over time. Based on the data and defined condition thresholds the component will also

estimate the remaining road life or the expected period before maintenance becomes necessary.

6. Maintenance Forecasting and Recommendation

The predicted future condition and remaining-life estimates will be converted into information that can support maintenance planning. Road sections approaching condition levels will be identified and preventive maintenance recommendations will be generated according to their predicted condition, deterioration pattern and maintenance requirement.

7. Model Validation and Evaluation

Model performance will be evaluated using test data. Regression predictions will be assessed using measures such as Mean Absolute Error (MAE) Root Mean Squared Error (RMSE) and R^2. Where the component performs classification, Precision, Recall and F1-score will be considered. The results of candidate models will be compared to select the approach that provides the reliable performance for the available dataset.

8. Integration with the Complete System

Finally the predictive component will be integrated with the road maintenance solution. It will receive road information, from the system and return future condition predictions, remaining road life estimates, maintenance forecasts and preventive maintenance recommendations. This allows the solution to support both current road assessment and forward-looking maintenance planning.

3.2 Data / Participants / Experimental Inputs

The study will mainly use road-related data to develop and test the predictive maintenance component. Historical records, supporting information, and a prototype dataset will be used during model development.

3.2.1 Purchase History Data

Historical road data will include previous road condition records, identified defects, maintenance history, road characteristics, and other available information related to road deterioration. These records will help the model learn how road conditions change over time.

3.2.2 Questionnaire Data

Additional information such as traffic conditions, rainfall, temperature, road age, and road type will be considered where available. These factors may help explain differences in deterioration between road sections.

3.2.3 Synthetic / Prototype Dataset

If sufficient real-world historical data is not available during the initial development stage, a prototype dataset will be created using realistic road-condition attributes and value ranges. This dataset will mainly be used to test the workflow and system integration before evaluation with suitable real-world data.

3.2.4 Data Preprocessing

Before model training, the data will be cleaned by handling missing values, duplicate records, inconsistent formats, and unsuitable values. Relevant features will then be selected and transformed into a suitable format for machine-learning models.

3.3 Baseline / Benchmark

A baseline will be created to see if the new machine-learning approach makes a difference in predicting what happens to roads in the future. Of looking at the new models on their own their results will be checked against simpler ways of making predictions.

For predicting road conditions and how long the roads will last a simple model based on regression will be used as the starting point. This gives an example to see if more advanced machine-learning methods can understand how roads get worse better. When the task is about sorting things into categories like deciding if a road needs work or what level of risk it has a simple model like Logistic Regression or a basic Decision Tree will be used as the baseline.

The results from the baseline will then be compared to models made using techniques like Random Forest, XGBoost, CatBoost and LightGBM. All models will use the way of preparing data and the same way of training and testing so that the comparison is fair.

For tasks where numbers are predicted the results will be measured using MAE, RMSE and R^2. For tasks where categories are predicted the results will be measured using Precision, Recall and F1-score. The best model will not be chosen based on one number; the models reliability, how consistent it is and how well it works with the road data available will also be considered.

This way of checking models will help show if the chosen model is much better, than options and if it is worth using in the part of the system that predicts road maintenance needs.

3.4 Evaluation Metrics

The performance of the developed models will be evaluated using the following metrics.

Prediction Task	Metric	Purpose
Future Road Condition	MAE	Measures the average prediction error.
Future Road Condition	RMSE	Measures prediction error while giving more weight to larger errors.
Future Road Condition	R^2 Score	Measures how well the model explains changes in the actual values.
Maintenance Need / Risk Level	Precision	Measures how accurately the model identifies roads that require attention.
Maintenance Need / Risk Level	Recall	Measures how well the model detects road sections that actually require attention.
Maintenance Need / Risk Level	F1-Score	Provides a balanced measure between Precision and Recall.

3.5 Validation Strategy

The predictive maintenance component will be tested using data that was not used when training the model. The dataset available will be split into training and testing parts. This helps to see how well the model works on road data it has never seen before.

When there is data, k-fold cross-validation will also be applied during model development. This method helps reduce the chance of picking a model that only does well on one set of data. Different machine-learning models will be tested under conditions so their results can be compared fairly.

The predicted future condition of roads and the remaining life of the roads will be compared with historical observations. Maintenance risk levels and recommendations will also be checked against predefined thresholds for road conditions. This ensures that the outputs make sense and are useful for planning maintenance activities.

The model that shows the consistent results throughout the validation steps will be chosen for the final implementation, in Component 4

3.6 Development Approach

Component 4 will be built using an iterative and incremental approach. The development will start with a data-processing and prediction workflow. That workflow will then be improved through repeated testing and evaluation.

The main development stages are:

1. **Data Preparation** – Gather, organise road-related data.

2**. Prototype Development** – Build a prediction model using selected road features.

3. **Model Experimentation** – Train and compare machine-learning models such as Random Forest, XGBoost, CatBoost and LightGBM.

4. **Prediction and Recommendation Development** – Generate road condition, remaining road life, maintenance forecasts and preventive recommendations.

5. **Testing and Improvement** – Evaluate model. Refine the selected approach.

6. **System Integration** – Component 4 with the other components of the complete road maintenance system.

This way Component 4 can be developed step by step. It allows the prediction model to improve based on test results and the data that becomes available, over time. The approach keeps things responsive to feedback.

3.7 Tools and Platforms with Justification

The following tools and technologies will be used to develop and evaluate Component 4.

Tool / Platform	**Proposed Purpose**	**Feasibility Justification**
Python	Main development language	Provides strong support for data analysis and machine-learning development.
Pandas	Data cleaning and preparation	Useful for handling structured road datasets, missing values, and data transformation.
NumPy	Numerical processing	Supports efficient numerical calculations required during data preparation and modelling.
Scikit-learn	Preprocessing, baseline models and evaluation	Provides commonly used ML algorithms, data splitting, cross-validation, and evaluation metrics.
XGBoost	Predictive model development	Suitable for learning complex patterns from structured road-related data..
CatBoost	Predictive model development	Useful when the dataset contains both numerical and categorical road attributes.
Git/GitHub	Version control	Helps to manage and track project code changes.

Table 1: Tools and Platforms with Justification

4. HIGH-LEVEL SYSTEM ARCHITECTURE

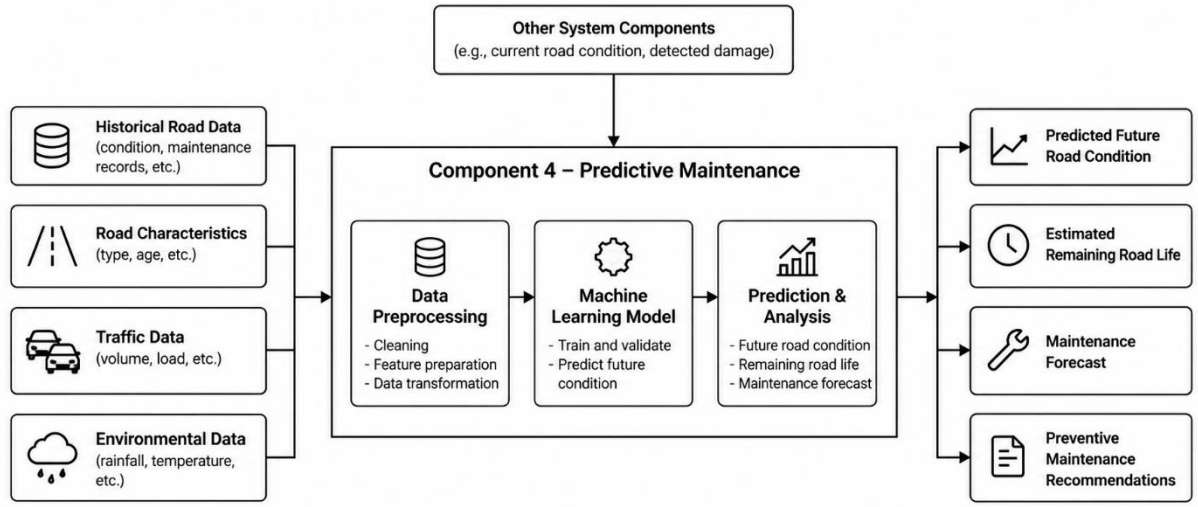
4.1 Overview of the Proposed Solution

Component 4 proposes a AI-based predictive maintenance solution that supports early road maintenance planning. Of focusing only on the current condition of a road the component uses current and historical road-related data to understand deterioration patterns and predict how road conditions may change over time.

The component will process information such as road condition, detected damage, maintenance history, road characteristics, traffic-related factors and environmental conditions where available. Machine-learning models will then be used to generate four outputs: future road condition, remaining road life, maintenance forecast and preventive maintenance recommendations.

Within the proposed system Component 4 acts, as the prediction and planning layer. It uses information generated by other components and converts it into forward-looking information that can help identify road sections that may require attention before severe deterioration occurs.

Road Data → Data Preprocessing → ML Prediction → Future Condition & Remaining Road Life → Maintenance Forecast → Preventive Recommendation



4.2. Individual Component Architecture

Component 4 follows a simple data-processing and prediction architecture. Road-related data is first collected from historical records and relevant outputs from other system components. The data is then cleaned and prepared before being passed to the selected machine-learning model.

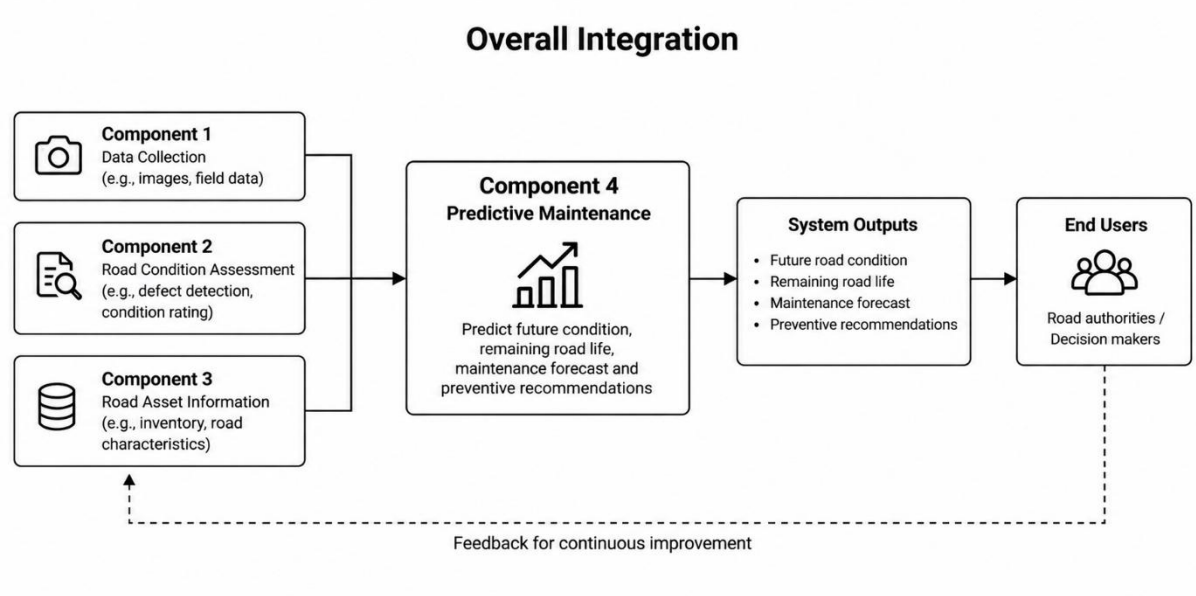
The model predicts future road conditions and estimates the remaining road life. These results are then used to forecast maintenance needs and provide preventive maintenance recommendations.

Flow:
Road Data → Data Preprocessing → ML Model → Future Condition Prediction → Remaining Road Life → Maintenance Forecast → Preventive Recommendation

4.3 Overall Integration

My component does not work independently.

The integration can be represented as:



4.4 Dependencies

Dependency	Purpose
Historical Road Data	Provides detected defects and damage information required for deterioration analysis.
Road Characteristics	Provides information such as road type, age, and other relevant attributes.
Traffic Data	Helps identify the effect of traffic conditions on road deterioration.
Environmental Data	Provides factors such as rainfall and temperature where available.
Maintenance History	Provides information about previous maintenance activities and their timing.
System Integration	Allows Component 4 to receive required inputs and provide predictions to the complete system.

4.5 Individual Responsibility Boundary

Individual Responsibility Boundary
Component-wise Activity Distribution

Activity / Deliverable	My Responsibility	Shared Responsibility	Other Member
Purchase-history data preparation	✓		
Shopping pattern analysis	✓		
Recommendation model	✓		
Monthly product prediction	✓		
Recommendation API	✓	✓	
Recommendation UI		✓	✓
Product recognition			✓
Trolley tracking			✓
Cart product detection		✓	✓
Billing synchronization	✓	✓	
Fiall system integration		✓	✓
Final system testing		✓	✓

5. USER REQUIREMENTS

5.1 Stakeholders

Component 4 is mainly used by road authorities, road engineers, maintenance teams, system administrators and decision makers.

Road authorities and road engineers will use predicted information, for maintenance planning while system administrators will support data management and system integration.

Other research components will also interact with Component 4 by providing road information.

5.2 Evidence Collection

The requirements for Component 4 will be identified using existing research studies, available road-condition data, historical maintenance information, and current road-maintenance practices. Requirements identified through discussions with supervisors and the integration needs of the other system components will also be considered.

5.3 Functional Requirement

The main functional requirements of Component 4 are:

Receive historical road-related data.

Preprocess the collected data.

Predict road conditions using machine learning.

Estimate the remaining road life.

Forecast future maintenance requirements.

Generate maintenance recommendations.

Share the prediction results with the road maintenance system.

5.4 Non-Functional Requirement

The component must deliver predictions that are reasonably accurate. It should also run efficiently without slowing down the system. As more road data comes in the system needs to scale up. It should be easy to update models when new data arrives so it remains relevant over time. All information related to roads must be managed securely protecting details. The final predictions and recommendations should be clear and straightforward. They need to help decision-makers with maintenance planning.

6.0 COMMERCIALIZATION PLAN

The proposed road maintenance system has the potential to be developed as a decision-support solution for organizations responsible for managing road networks. The system can provide road-condition information, future deterioration predictions, maintenance forecasts, and preventive maintenance recommendations through a single platform.

6.1 Target Market

The main target market includes government road authorities, municipal councils, provincial road agencies, engineering organizations, and private companies involved in road construction and maintenance. The solution may initially focus on organizations managing medium to large road networks.

6.2 Customer Persona

A typical customer would be a road engineer, maintenance manager, or infrastructure planner responsible for monitoring road conditions and planning maintenance activities.

Main

Problems:

Maintenance decisions can be delayed because road-condition information is scattered or only considered after visible damage occurs. It can also be difficult to identify which roads may require attention in the near future.

Customer

Need:

Users need a simple way to understand current road conditions, identify future deterioration risks, and plan maintenance earlier.

Proposed

Solution:

The proposed system combines road-condition information with AI-based predictions to provide future road condition, remaining road life, maintenance forecasts, and preventive maintenance recommendations.

6.3 Value Proposition

The main value of the proposed solution is its ability to move road maintenance from a mainly reactive process towards a more predictive and preventive approach. Instead of only showing existing road damage, the system provides information that can support earlier maintenance decisions.

6.4 Revenue Model

A combination of deployment, subscription, maintenance, and analytics services can be considered when commercializing the system.

6.4.1 System Deployment Fee

An initial fee can be charged for system configuration, integration, data preparation, and deployment according to the requirements of each organization.

6.4.2 Software Subscription

Organizations can pay a monthly or annual subscription to access the road monitoring, prediction, and maintenance-planning platform.

6.4.3 Maintenance and Support

Additional support packages can cover system maintenance, technical assistance, model updates, and software improvements.

6.4.4 Analytics Service

Advanced reports and road deterioration analysis can be provided as an additional service for organizations requiring more detailed maintenance-planning information.

6.4.5 Proposed Revenue Structure

The main revenue can come from initial deployment fees and recurring software subscriptions, while maintenance, support, and advanced analytics can provide additional income.

6.5 Cost Estimation

The main costs are expected to include software development, cloud or server resources, data collection and storage, model development, system integration, testing, maintenance, and technical support. Actual costs will depend on the scale of deployment and amount of road data processed.

6.6 Pricing Strategy

A tiered pricing strategy can be used based on the size of the road network, number of users, data volume, and features required. Smaller organizations can use a basic package, while larger authorities can use advanced prediction and analytics features.

6.7 Competitive Advantage

The main competitive advantage is the integration of current road assessment with future-oriented maintenance support. Component 4 adds value by predicting future condition and remaining road life and converting these predictions into maintenance forecasts and preventive recommendations.

6.8 Intellectual Property Considerations

The developed prediction workflow, trained models, recommendation logic, software code, system design, and documentation can be treated as project intellectual property where applicable. Third-party datasets, libraries, and machine-learning frameworks will be used according to their respective licences. Any future commercialization should also consider data ownership, software licensing, and protection of project-developed assets.

7.0 BUDGET AND JUSTIFICATION

7.1 Hardware costs

Hardware	Justification	Estimated Cost
Existing Laptop / Computer	Personal computer will be used for development and testing.	0
Cloud Computing Resources	May be required for model training and experiments if local resources are insufficient.	LKR 10,000
Data Storage / Backup	Used for storing datasets, trained models, and backups.	LKR 5,000
Estimated Hardware Total		**LKR 15,000**

Table 2: Hardware costs

7.2 Software Tools

Tool / Platform	**Purpose**	**Cost**
Python	ML and data processing	Free
Pandas	Dataset processing	Free
Google Colab	Cloud GPU environment	Free
NumPy	Numerical processing	Free
XGBoost / CatBoost / LightGBM	Backend API	Free
Jupyter Notebook / Google Colab	User interface and mobile application	Free
Git/GitHub	Version control	Free

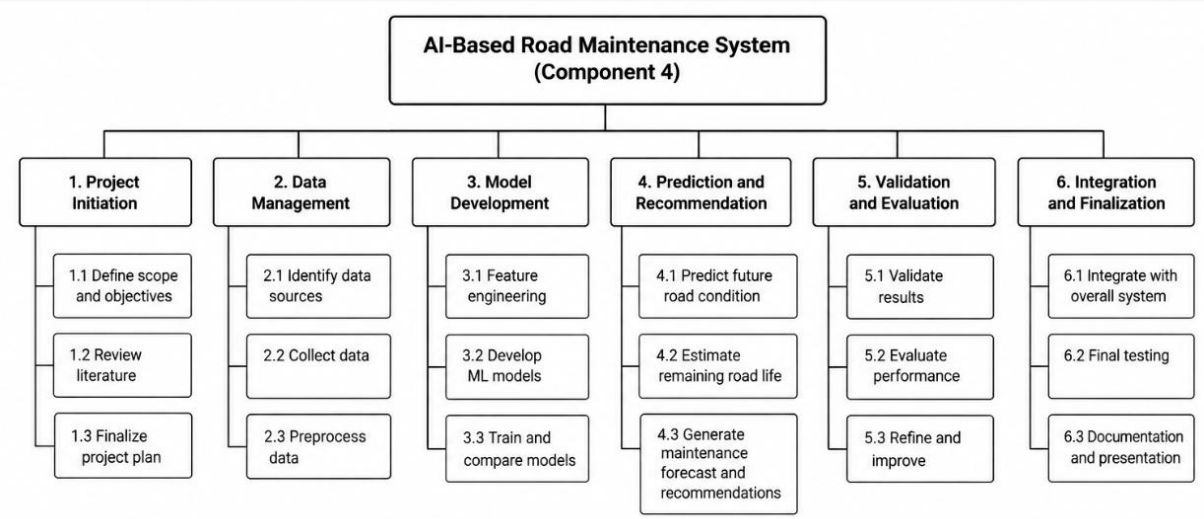
Table 3 : Software Tools

7.3 Human Effort Cost

Activity	Description
Data Collection	Preparing purchase-history data
Data Preprocessing	Cleaning and organizing data
ML Development	Training and testing recommendation model
API Development	Connecting model with system
Testing	Model and system evaluation
Documentation	Report and research documentation

Table 4 : Human Effort Cost

8.0 WORK BREAKDOWN STRUCTURE



The Work Breakdown Structure divides the development of Component 4 into six stages.

1. Project Initiation: Defines the research problem, scope, objectives and overall plan for the maintenance component.

2. Data Management: Identifies required road data collects datasets and preprocesses the data for model development.

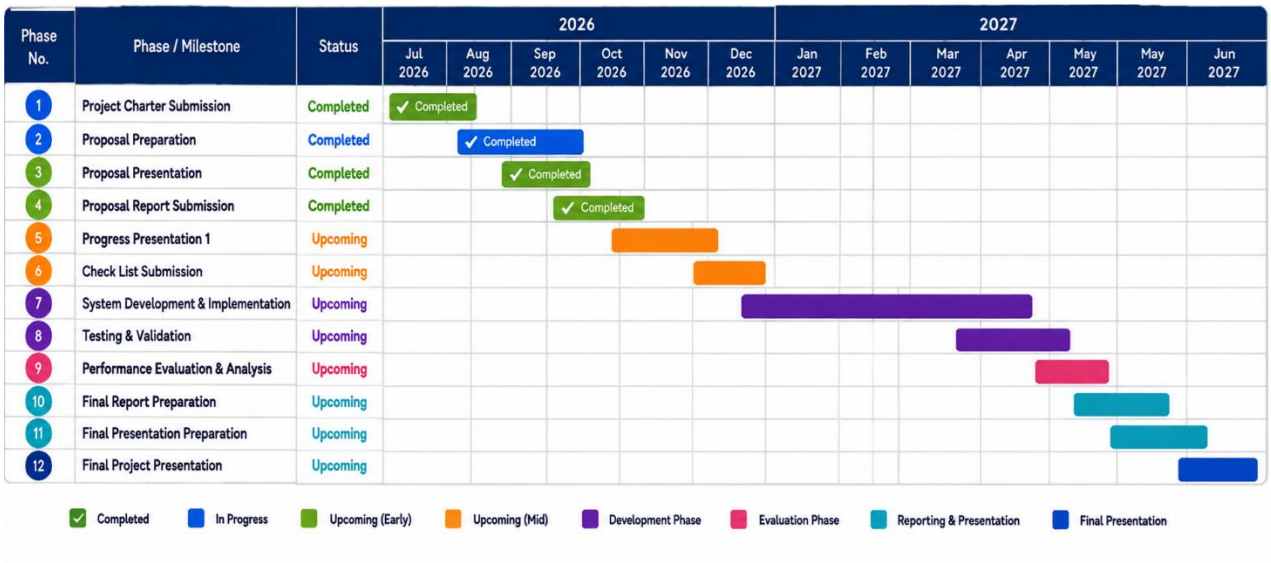
3. Model Development: Selects features and develops, trains and compares suitable machine-learning models.

4. Prediction and Recommendation: Predicts road condition, estimates remaining road life forecasts maintenance needs and provides preventive maintenance recommendations.

5. Validation and Evaluation: Evaluates the developed models using selected performance metrics. Improves the model where necessary.

6. Finalization: Integrates Component 4, with the road maintenance system performs final testing and completes the documentation.

9.0 GANTT CHART



Milestone 1 – Research Problem and Scope Finalized

Milestone 2 – Literature Review and Research Gap Identified

Milestone 3 – Requirements and Data Sources Identified

Milestone 4 – Dataset Collection and Preparation Completed

Milestone 5 – Data Preprocessing and Feature Engineering Completed

Milestone 6 – Initial ML Model Developed

Milestone 7 – ML Models Trained and Compared

Milestone 8 – Future Road Condition and Remaining Road Life Prediction Completed

Milestone 9 – Maintenance Forecasting and Recommendation Module Completed

Milestone 10 – Model Validation and Evaluation Completed

Milestone 11 – Component 4 Integrated with Overall System

Milestone 12 – Final Testing, Documentation and Research Completion

10. REFERENCES LIST

[1] M. S. Bin Aamir, M. H. Ilyas, F. Khalique, B. Bashir, S. Mahmood, and R. A. Khalil, "Autonomous Road Defects Segmentation UsingTransformer-Based Deep Learning Models With Custom Dataset," *IEEE Access*, vol. 13, pp. 174177–174199, 2025, doi:10.1109/ACCESS.2025.3617336.

[2] Y. Song, Y. D. Wang, X. Hu, and J. Liu, "An Efficient and Explainable Ensemble Learning Model for Asphalt Pavement ConditionPrediction Based on LTPP Dataset," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22084–22093, Nov. 2022, doi:10.1109/TITS.2022.3164596.

[3] N. S. Ahmed, "Machine Learning Models for Pavement Structural Condition Prediction: A Comparative Study of Random Forest (RF)and eXtreme Gradient Boosting (XGBoost)," *Open J. Civil Eng.*, vol. 14, pp. 570–586, 2024, doi: 10.4236/ojce.2024.144031.

[4] J. Ou, J. Zhang, H. Li, and B. Duan, "Road damage prediction and intelligent maintenance methods based on stacking ensemblelearning," *Advanced Engineering Informatics*, vol. 66, Art. no. 103466, 2025, doi: 10.1016/j.aei.2025.103466.

[5] E. I. Obanor, A. O. Adeodu, D. E. Ighrawve, J. C. Nkwor, and O. I. Richard, "AI-Based Decision Support Systems in Project Management:Enhancing Efficiency and Accuracy," *INCODING: Journal of Informatic and Computer Science Engineering*, vol. 5, no. 2, pp. 341–352, 2025,doi: 10.34007/incoding.v5i2.993.

[6] "Predicting Pavement Structural Condition Using Machine Learning Methods," *Sustainability*, vol. 14, no. 14, Art. no. 8627, 2022.

[7] "A maintenance-aware machine learning framework for network-level highway pavement condition prediction," *Computer-Aided CivilandInfrastructure Engineering*, vol. 47, Art. no. 100091, 2026, doi: 10.1016/j.cacaie.2026.100091.

[8] "Explainable AI for predicting pavement roughness under maintenance and no-maintenance scenarios," 2025. The study compares ANN,Random Forest, XGBoost, and CatBoost for multi-year pavement-condition prediction and examines traffic and structural features as predictors.

[9] "A software for predicting Pavement Condition Index (PCI) using machine learning for practical decision-making with an exclusion approach,"2025. The study applies Random Forest and XGBoost using factors including road age, pavement characteristics, traffic and distress information.

[10] Y. Song, Y. D. Wang, X. Hu, and J. Liu, “An Efficient and Explainable Ensemble Learning Model for Asphalt Pavement Condition Prediction Basedon LTPP Dataset,” *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22084–22093, 2022.

11. APPENDICES

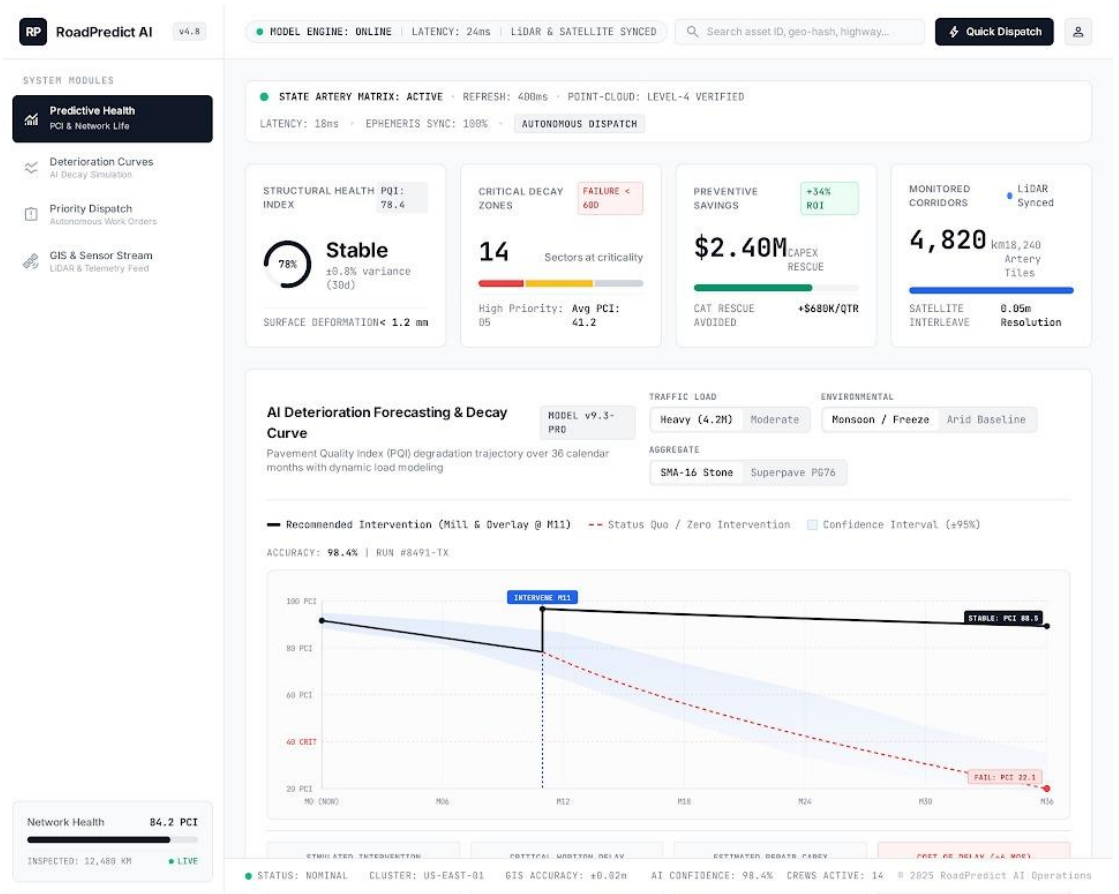
Appendix A – Proposal Evidence and Decision Log

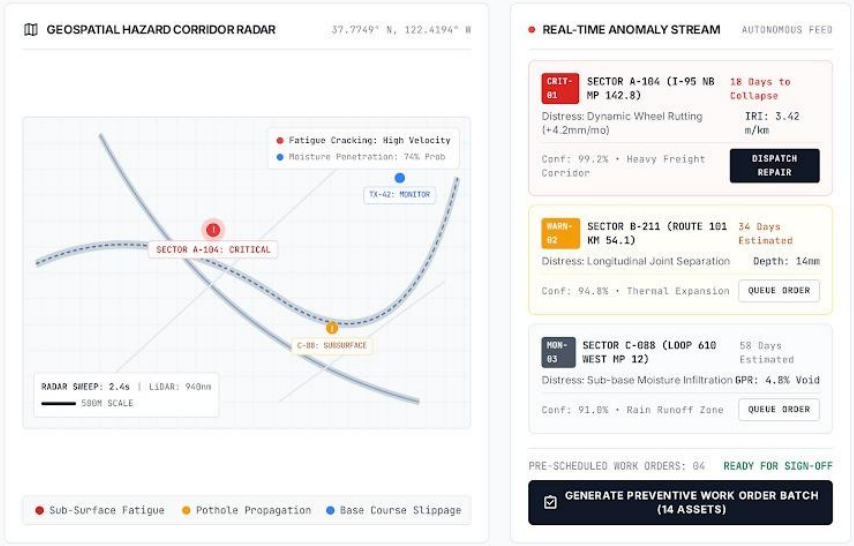
Date	Claim / Problem	Evidence Collected	Alternatives Considered	Decision Made	Supervisor Discussion

Appendix 3 — AI Use Disclosure

AI Tool	Purpose of Use	What Was Generated/Assisted	How Output Was Verified	Final Student Contribution
Grok (xAI)	Drafting support & structure alignment	Initial proposal structure and literature synthesis	Cross-checked against TAF, papers & template	All technical decisions, gap statement objectives & evaluation design owned by student

Using AI does not imply that the responsibility for the submitted work is transferred. The student is solely accountable for accuracy, originality, validity of references, research decision, methodology, technical decisions, interpretation of evidence, and defensibility of all aspects of the proposal.


RoadPredict AI
v4.8
MODEL ENGINE: ONLINE | LATENCY: 24ms | LiDAR & SATELLITE SYNCED
Search asset ID, geo-hash, highway...
Quick Dispatch
SYSTEM MODULES
Predictive Health
PCI & Network Life
Deterioration Curves
AI Decay Simulation
Priority Dispatch
Autonomous Work Orders
GIS & Sensor Stream
LiDAR & Telemetry Feed
STATE ARTERY MATRIX: ACTIVE · REFRESH: 400ms · POINT-CLOUD: LEVEL-4 VERIFIED
LATENCY: 18ms · EPHEMERIS SYNC: 100% · AUTONOMOUS DISPATCH
STRUCTURAL HEALTH INDEX
PQI: 78.4
78%
Stable
±0.8% variance (30d)
SURFACE DEFORMATION
< 1.2 mm
CRITICAL DECAY ZONES
FAILURE < 60D
14
Sectors at criticality
High Priority: 05
Avg PCI: 41.2
PREVENTIVE SAVINGS
+34% ROI
$2.40M
CAPEX RESCUE
CAT RESCUE AVOIDED
+$680K/QTR
MONITORED CORRIDORS
LiDAR Synced
4,820
km
18,240 Artery Tiles
SATELLITE INTERLEAVE
0.05m Resolution
AI Deterioration Forecasting & Decay Curve
MODEL v9.3-PRD
Pavement Quality Index (PQI) degradation trajectory over 36 calendar months with dynamic load modeling
TRAFFIC LOAD
Heavy (4.2M)
Moderate
ENVIRONMENTAL
Monsoon / Freeze
Arid Baseline
AGGREGATE
SMA-16 Stone
Superpave PG76
Recommended Intervention (Mill & Overlay @ M11)
Status Quo / Zero Intervention
Confidence Interval (±95%)
ACCURACY: 98.4% | RUN #8491-TX
INTERVENE M11
STABLE: PCI 88.5
FAIL: PCI 22.1
100 PCI
80 PCI
60 PCI
40 CRIT
20 PCI
M0 (NOW)
M6
M12
M18
M24
M30
M36
Network Health
84.2 PCI
INSPECTED: 12,480 KM
LIVE
STATUS: NOMINAL
CLUSTER: US-EAST-01
GIS ACCURACY: ±0.02m
AI CONFIDENCE: 98.4%
CREWS ACTIVE: 14
© 2025 RoadPredict AI Operations


GEOSPATIAL HAZARD CORRIDOR RADAR
37.7749° N, 122.4194° W
Fatigue Cracking: High Velocity
Moisture Penetration: 74% Prob
TX-42: MONITOR
SECTOR A-104: CRITICAL
C-88: SUBSURFACE
RADAR SWEEP: 2.4s | LiDAR: 940nm
500M SCALE
Sub-Surface Fatigue
Pothole Propagation
Base Course Slippage
REAL-TIME ANOMALY STREAM
AUTONOMOUS FEED
CRIT-01
SECTOR A-104 (I-95 NB MP 142.8)
18 Days to Collapse
Distress: Dynamic Wheel Rutting (+4.2mm/mo)
IRI: 3.42 m/km
Conf: 99.2% · Heavy Freight Corridor
DISPATCH REPAIR
WARN-02
SECTOR B-211 (ROUTE 101 KM 54.1)
34 Days Estimated
Distress: Longitudinal Joint Separation
Depth: 14mm
Conf: 94.8% · Thermal Expansion
QUEUE ORDER
MON-03
SECTOR C-088 (LOOP 610 WEST MP 12)
58 Days Estimated
Distress: Sub-base Moisture Infiltration
GPR: 4.8% Void
Conf: 91.0% · Rain Runoff Zone
QUEUE ORDER
PRE-SCHEDULED WORK ORDERS: 04
READY FOR SIGN-OFF
GENERATE PREVENTIVE WORK ORDER BATCH (14 ASSETS)


TARGET SECTOR STRUCTURAL DIAGNOSTICS & TELEMETRY LEDGER
Total Inspected: 1,482 Corridor Segments
ASSET ID
CORRIDOR NAME
SURFACE LAYER
PCI CURRENT
IRI ROUGHNESS
PREDICTED DECAY
RECOMMENDED MITIGATION
ACTION
SEC-9904-TX
Interstate 35 Express (NB Lane 1)
SMA Stone Matrix (2021)
42.1 [CRIT]
3.88 m/km
-1.8 PCI/mo
Cold In-Place Recycling (CIR)
EXECUTE
SEC-8812-FL
Turnpike Mainline MP 88.4
Dense-Graded Superpave
61.5 [WARN]
2.10 m/km
-0.6 PCI/mo
Micro-surfacing + Crack Seal
SCHEDULE
SEC-3301-CA
Highway 101 South Coastal Artery
Porous Friction Course
82.4 [NOM]
1.02 m/km
-0.2 PCI/mo
Routine Slurry Seal Cycle
MONITOR

RoadPredict AI
INFRA-ENGINE
v4.8

Deterioration Forecasting & Neural Vector

ENGINE: ONLINE | LATENCY: 24ms | SATELLITE + LiDAR SYNC

Search asset, geo-hash, route

Quick Dispatch

SYSTEM MODULES

- Predictive Health — PCI & Network Life
- Deterioration Curves — AI Decay Simulation
- Priority Dispatch — Autonomous Work Orders
- GIS & Sensor Stream — LiDAR & Telemetry Feed

NETWORK HEALTH 84.2 PCI

INSPECTED: 12,480 KM ● LIVE

SIMULATION ENGINE v3.4α STOCHASTIC DECAY MODELER

Asset Target: US-101-SEC_0842-N

RADAR ENSEMBLE: 54k PTS

RUN NEURAL DEGRADATION MODEL

FORECAST HORIZON: 30D | 90D | 180D | 1Y | 3Y

PRECIPITATION SURGE +45.0% — 720mm

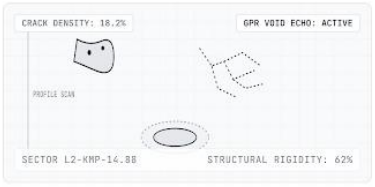
AXLE STRESS (ESAL) 12k HDV/D — 1.84μ

THERMAL DELTA Δ 17°C — MIN: 21°C → MAX: 38°C

T-0 Baseline Observation

PCI: 68.4 / MODERATE

SENSOR OVERLAY: LIDAR + GPR 800MHz DEPTH: -0.45m TO SURFACE



POTHOLE SEEDS	FATIGUE AREA	CAVITY VOID
07	18.2%	0.34m³
Dia < 65mm	Wheel-path Zone	GPR Radar Refl

RECOMMENDED INTERVENTION: SLURRY SEAL + CRACK INJECTION

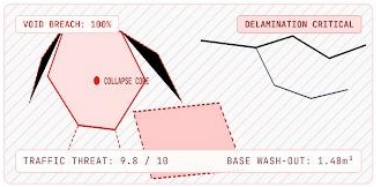
PROACTIVE MITIGATION COST $6,200

ACTION WINDOW ≤ 21 DAYS REMAINING

T+90D AI Unmitigated Projection

PCI: 31.0 / FAILED

SIMULATION: MONTE CARLO SEVERE WEATHER EROSION FACTOR: 3.42X



FAILURE PROBABILITY	RAVELING LOSS	STRUCTURAL RISK
96.8%	68.4mm	CAT-04
High-Impact Event	Binder Depleted	Foundation Flaw

MANDATORY RECONSTRUCTION: FULL BASE STABILIZATION + REPAVE

EMERGENCY REBUILD COST $48,000

NET DEFERRAL LOSS +$41,800 (+674%)

Pavement Stratigraphy & Geotechnical Vulnerability

SCAN MODE: DUAL POLARIZATION GPR · CALIBRATION: ASTM D6432

STRATIGRAPHIC ELEVATION SCHEMATIC — MAGNIFICATION: 3.5X

L1: WEARING COURSE (ASPHALT CONCRETE) Thickness: 50mm | Dense-Graded Superpave	Shear: 480 kPa	SAT: 94.2%
L2: BINDER COURSE (INTERMEDIATE MIX) Thickness: 90mm | Aggregate Interlock 19mm	Shear: 310 kPa	SAT: 78.5%
L3: CRUSHED STONE BASE COURSE Thickness: 160mm | Crushed Granite CRB-1	Shear: 190 kPa	SAT: 61.3%
L4: COMPACTED SUBGRADE (NATIVE SILT/CLAY) Depth: Infinite (>300mm tested) | CBR: 4.8%	Pore Pressure: +42 kPa	SAT: 42.0%

PERCOLATION INGRESS RATE: 14.2 mm/hr — INTERFACE FRICTION: 0.72

L1 Surface Course Strain — LIMIT: 1200 με
1,840 — THRESHOLD BREACHED +53.3% OVERSTRESSED

Subsurface Moisture Pumping Index — TOLERANCE: 50%
82.1 — HYDRAULIC CRACKING RISK — HIGH CAVITATION

L3 Resilient Modulus (Mr) — TARGET: > 180 MPa
112 MPa — SUB-OPTIMAL ELASTICITY -37.7% DEGRADED

GEOTECHNICAL ADVISORY: BASE DRAINAGE COLLAPSE IMMINENT

Probabilistic Network Failure Matrix · Top Critical Corridors

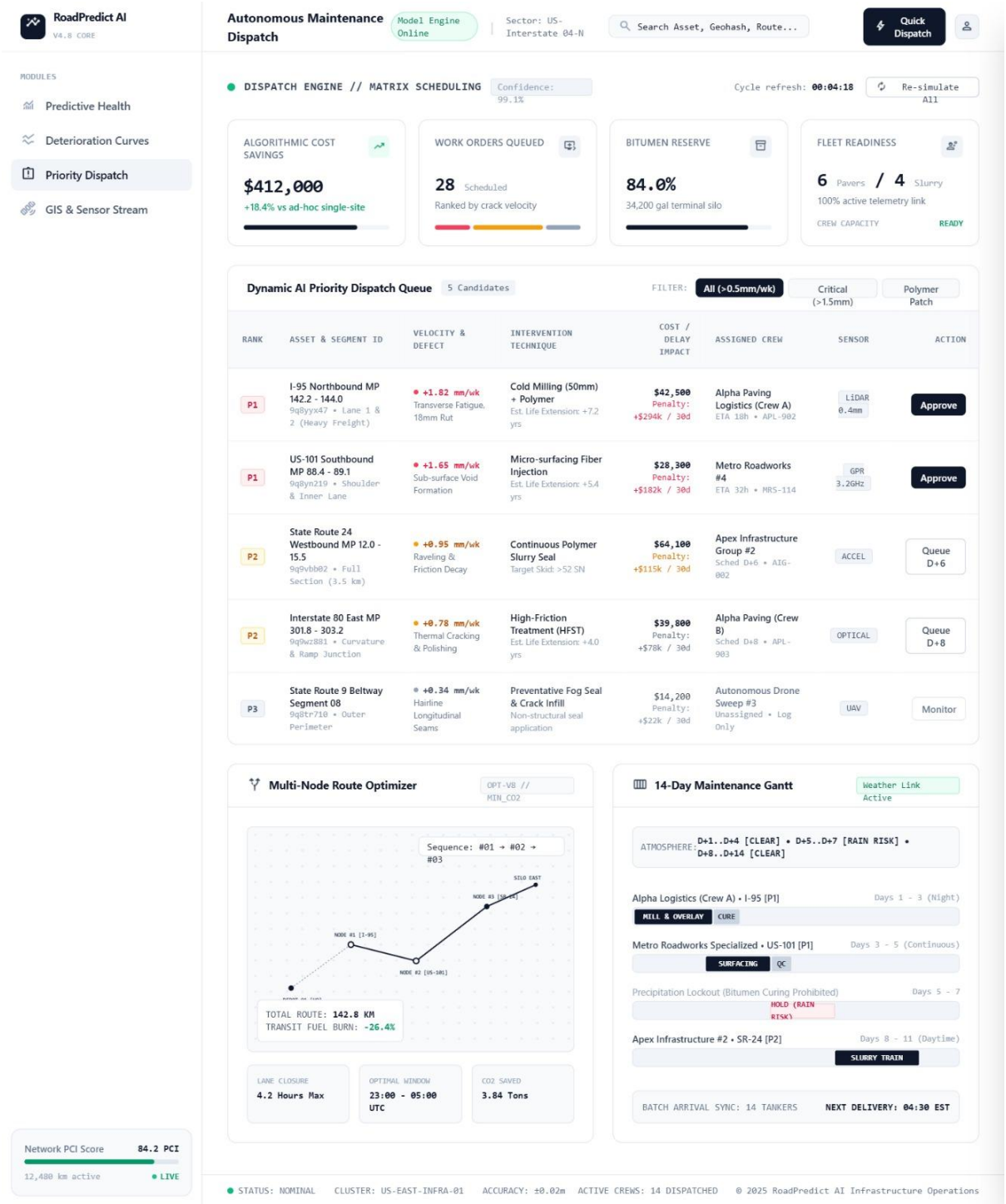
EXPORT CSV | QUEUE DISPATCH ALL

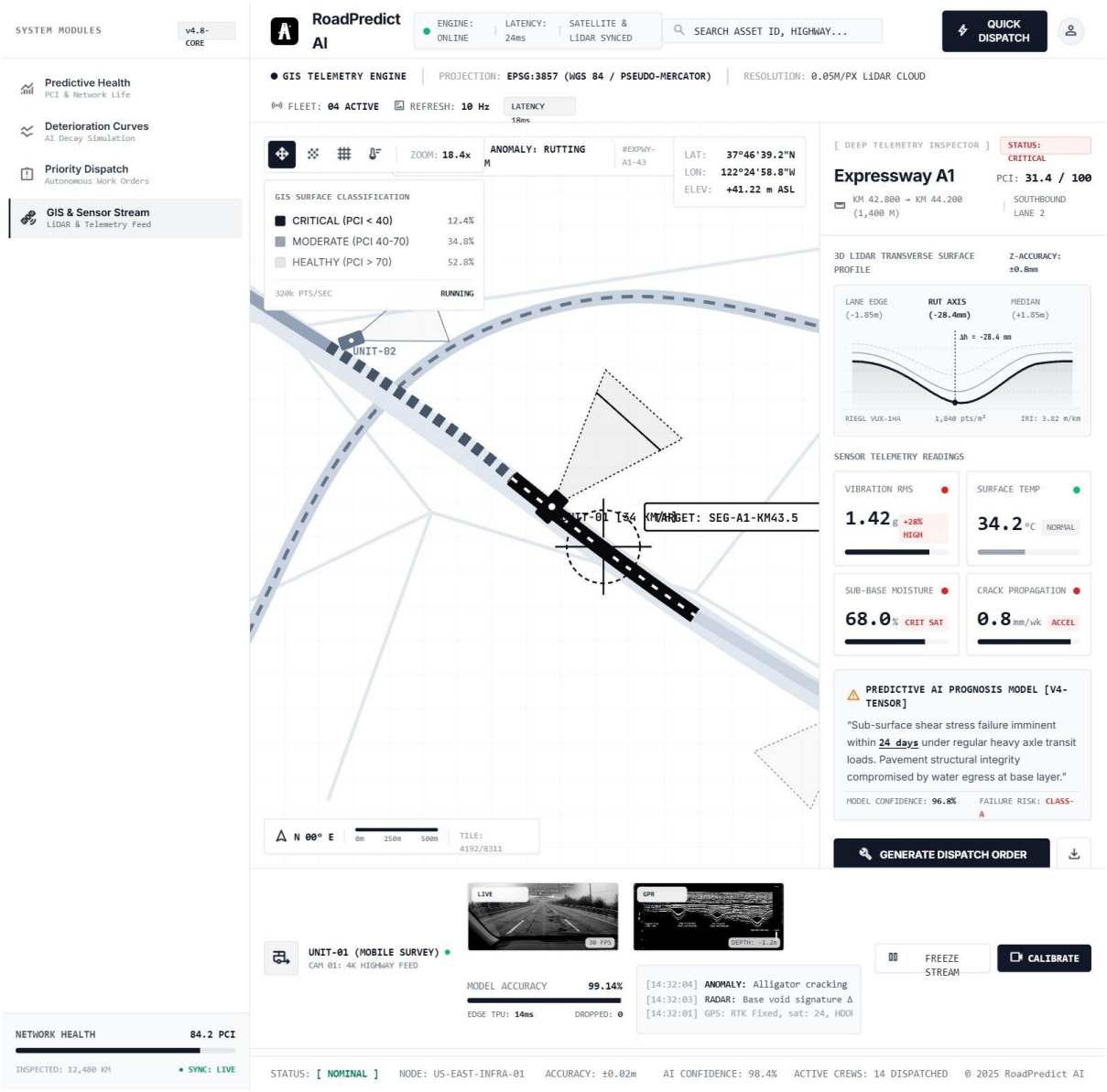
SEGMENT ID	COORDINATES / MILEPOST	CURRENT PQI	PROJECTED FAILURE	FAILURE MECHANISM	REC. INTERVENTION	ACTION
US-101-SEC_0842-N	37.7749° N, 122.4194° W (MP 14.88)	54.2	T+18 DAYS	THERMAL CRACK PROPAGATION	Crack Sealing + High Modulus Seal	DISPATCH
I-80-SEC_1209-E	38.5816° N, 121.4944° W (MP 92.14)	48.0	T+27 DAYS	RUTTING DEPTH (>18mm)	Mill & Inlay (Wearing 50mm)	DISPATCH
CA-99-SEC_0441-S	36.7468° N, 119.7726° W (MP 44.02)	61.5	T+41 DAYS	SHEAR SLIPPAGE DELAMINATION	Microsurfacing Type III	DISPATCH
SR-17-SEC_0228-W	37.1124° N, 121.9840° W (MP 06.50)	39.8	T+52 DAYS	HYDRAULIC BASE EROSION	Full Depth Reclamation (FDR)	DISPATCH
I-580-SEC_3310-E	37.6888° N, 121.7680° W (MP 33.10)	63.0	T+76 DAYS	FATIGUE BLOCK CRACKING	Slurry Seal & Binder Replenishment	DISPATCH

TOP 5 HIGHEST DIVERGENCE NODES OF 384 MONITORED SECTORS — CONFIDENCE INTERVAL: 95% (BAYESIAN REGRESSION)

SYS.STATUS: NOMINAL | CLUSTER: US-EAST-INFRA-01 | GIS ACCURACY: ±0.02m | AI CONFIDENCE: 98.4% | REPAIR CREWS: 14 ACTIVE

© 2025 RoadPredict AI


RoadPredict AI
V4.8 CORE
MODULES
Predictive Health
Deterioration Curves
Priority Dispatch
GIS & Sensor Stream
Autonomous Maintenance Dispatch
Model Engine Online
Sector: US-Interstate 04-N
Search Asset, Geohash, Route...
Quick Dispatch
DISPATCH ENGINE // MATRIX SCHEDULING
Confidence: 99.1%
Cycle refresh: 00:04:18
Re-simulate All
ALGORITHMIC COST SAVINGS
$412,000
+18.4% vs ad-hoc single-site
WORK ORDERS QUEUED
28 Scheduled
Ranked by crack velocity
BITUMEN RESERVE
84.0%
34,200 gal terminal silo
FLEET READINESS
6 Pavers / 4 Slurry
100% active telemetry link
CREW CAPACITY READY
Dynamic AI Priority Dispatch Queue
5 Candidates
FILTER: All (>0.5mm/wk) Critical (>1.5mm) Polymer Patch
RANK
ASSET & SEGMENT ID
VELOCITY & DEFECT
INTERVENTION TECHNIQUE
COST / DELAY IMPACT
ASSIGNED CREW
SENSOR
ACTION
P1
I-95 Northbound MP 142.2 - 144.0
9q8yyx47 • Lane 1 & 2 (Heavy Freight)
+1.82 mm/wk
Transverse Fatigue, 18mm Rut
Cold Milling (50mm) + Polymer
Est. Life Extension: +7.2 yrs
$42,500
Penalty: +$294k / 30d
Alpha Paving Logistics (Crew A)
ETA 18h • APL-902
LiDAR 0.4mm
Approve
P1
US-101 Southbound MP 88.4 - 89.1
9q8yn219 • Shoulder & Inner Lane
+1.65 mm/wk
Sub-surface Void Formation
Micro-surfacing Fiber Injection
Est. Life Extension: +5.4 yrs
$28,300
Penalty: +$182k / 30d
Metro Roadworks #4
ETA 32h • MRS-114
GPR 3.2GHz
Approve
P2
State Route 24 Westbound MP 12.0 - 15.5
9q9vbb02 • Full Section (3.5 km)
+0.95 mm/wk
Raveling & Friction Decay
Continuous Polymer Slurry Seal
Target Skid: >52 SN
$64,100
Penalty: +$115k / 30d
Apex Infrastructure Group #2
Sched D+6 • AIG-002
ACCEL
Queue D+6
P2
Interstate 80 East MP 301.8 - 303.2
9q9wz881 • Curvature & Ramp Junction
+0.78 mm/wk
Thermal Cracking & Polishing
High-Friction Treatment (HFST)
Est. Life Extension: +4.0 yrs
$39,800
Penalty: +$78k / 30d
Alpha Paving (Crew B)
Sched D+8 • APL-903
OPTICAL
Queue D+8
P3
State Route 9 Beltway Segment 08
9q8tr710 • Outer Perimeter
+0.34 mm/wk
Hairline Longitudinal Seams
Preventative Fog Seal & Crack Infill
Non-structural seal application
$14,200
Penalty: +$22k / 30d
Autonomous Drone Sweep #3
Unassigned • Log Only
UAV
Monitor
Multi-Node Route Optimizer
OPT-V8 // MIN_CO2
Sequence: #01 → #02 → #03
SILO EAST
NODE #1 [I-95]
NODE #2 [US-101]
TOTAL ROUTE: 142.8 KM
TRANSIT FUEL BURN: -26.4%
LANE CLOSURE
4.2 Hours Max
OPTIMAL WINDOW
23:00 - 05:00 UTC
CO2 SAVED
3.84 Tons
14-Day Maintenance Gantt
Weather Link Active
ATMOSPHERE: D+1..D+4 [CLEAR] • D+5..D+7 [RAIN RISK] • D+8..D+14 [CLEAR]
Alpha Logistics (Crew A) • I-95 [P1]
Days 1 - 3 (Night)
MILL & OVERLAY
CURE
Metro Roadworks Specialized • US-101 [P1]
Days 3 - 5 (Continuous)
SURFACING
QC
Precipitation Lockout (Bitumen Curing Prohibited)
Days 5 - 7
HOLD (RAIN RISK)
Apex Infrastructure #2 • SR-24 [P2]
Days 8 - 11 (Daytime)
SLURRY TRAIN
BATCH ARRIVAL SYNC: 14 TANKERS
NEXT DELIVERY: 04:30 EST
Network PCI Score 84.2 PCI
12,480 km active
LIVE
STATUS: NOMINAL
CLUSTER: US-EAST-INFRA-01
ACCURACY: ±0.02m
ACTIVE CREWS: 14 DISPATCHED
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SYSTEM MODULES
v4.8-CORE
Predictive Health
PCI & Network Life
Deterioration Curves
AI Decay Simulation
Priority Dispatch
Autonomous Work Orders
GIS & Sensor Stream
LiDAR & Telemetry Feed
RoadPredict AI
ENGINE: ONLINE
LATENCY: 24ms
SATELLITE & LiDAR SYNCED
SEARCH ASSET ID, HIGHWAY...
QUICK DISPATCH
GIS TELEMETRY ENGINE
PROJECTION: EPSG:3857 (WGS 84 / PSEUDO-MERCATOR)
RESOLUTION: 0.05M/PX LiDAR CLOUD
FLEET: 04 ACTIVE
REFRESH: 10 Hz
LATENCY 18ms
ZOOM: 18.4x
ANOMALY: RUTTING
#EXPWY-A1-43
LAT: 37°46'39.2"N
LON: 122°24'58.8"W
ELEV: +41.22 m ASL
GIS SURFACE CLASSIFICATION
CRITICAL (PCI < 40) 12.4%
MODERATE (PCI 40-70) 34.8%
HEALTHY (PCI > 70) 52.8%
320k PTS/SEC
RUNNING
UNIT-02
UNIT-01
TARGET: SEG-A1-KM43.5
N 00° E
0m 250m 500m
TILE: 4192/8311
[DEEP TELEMETRY INSPECTOR]
STATUS: CRITICAL
Expressway A1
PCI: 31.4 / 100
KM 42.800 → KM 44.200 (1,400 M)
SOUTHBOUND LANE 2
3D LiDAR TRANSVERSE SURFACE PROFILE
Z-ACCURACY: ±0.8mm
LANE EDGE (-1.85m)
RUT AXIS (-28.4mm)
MEDIAN (+1.85m)
Δh = -28.4 mm
RIEGL VUX-1HA
1,840 pts/m²
IRI: 3.82 m/km
SENSOR TELEMETRY READINGS
VIBRATION RMS
1.42 g +28% HIGH
SURFACE TEMP
34.2 °C NORMAL
SUB-BASE MOISTURE
68.0% CRIT SAT
CRACK PROPAGATION
0.8 mm/wk ACCEL
PREDICTIVE AI PROGNOSIS MODEL [V4-TENSOR]
"Sub-surface shear stress failure imminent within 24 days under regular heavy axle transit loads. Pavement structural integrity compromised by water egress at base layer."
MODEL CONFIDENCE: 96.8%
FAILURE RISK: CLASS-A
GENERATE DISPATCH ORDER
LIVE
30 FPS
GPR
DEPTH: -1.2m
UNIT-01 (MOBILE SURVEY)
CAM 01: 4K HIGHWAY FEED
MODEL ACCURACY 99.14%
EDGE TPU: 14ms
DROPPED: 0
[14:32:04] ANOMALY: Alligator cracking
[14:32:03] RADAR: Base void signature Δ
[14:32:01] GPS: RTK Fixed, sat: 24, HDOI
FREEZE STREAM
CALIBRATE
NETWORK HEALTH 84.2 PCI
INSPECTED: 12,480 KM
• SYNC: LIVE
STATUS: [NOMINAL]
NODE: US-EAST-INFRA-01
ACCURACY: ±0.02m
AI CONFIDENCE: 98.4%
ACTIVE CREWS: 14 DISPATCHED
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